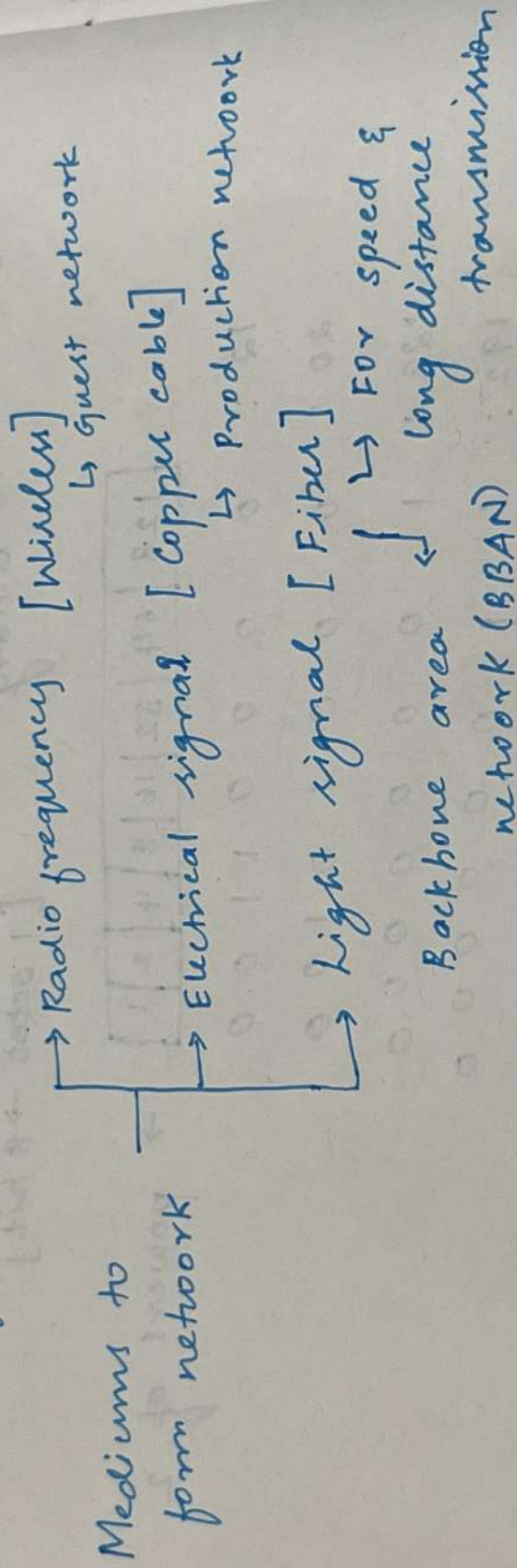


DAY-2 Physical Connections

NETWORK:

Two or more devices interconnected with each other for sharing resources, information & communication.



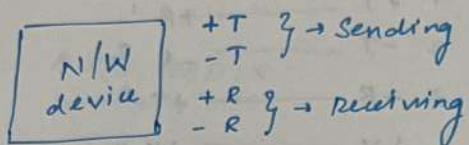
COPPER WIRES → Medium

UTP: Unshielded twisted Pair

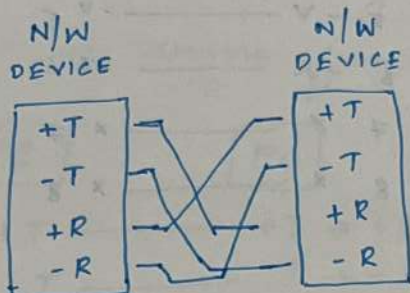
Contain 8 copper wires [4 pairs]

* Cable is a collection of wires

* A device which uses network to transmit or receive data is called as network device



* 4 wires are used to form a single network



* Crimping tool is used to connect a RJ45 Jack to a wire



* Process of connecting the wires to jack is called as crimping method. There are two types:

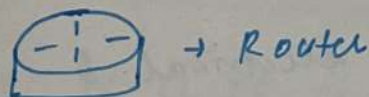
- 1) Straight cable
- 2) Cross cable

* Whether to use straight or cross cable depends on the functionality of device

DEVICE	+T	-T	+R	-R
PC	1	2	3	6
Router	1	2	3	6
Switch	3	6	1	2

* POE - Power ^{over} Ethernet
Transferring power via UTP cable. Eg: CCTV

* PC & Router are same function devices because it uses same pins.



→ Router



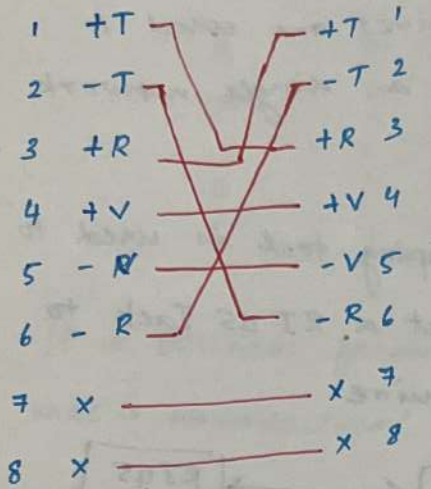
→ Switch

* 1, 2, 3, 6 are pins & there are 8 pins totally.

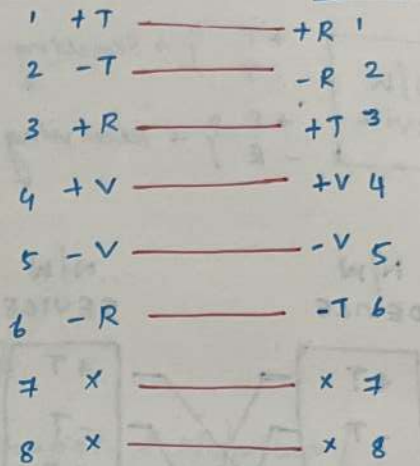
→ When there is a POE camera & non-POE switch, POE injector is used.

Same function device
Cross cable

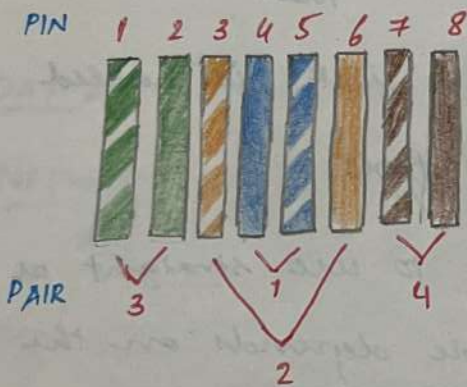
PC



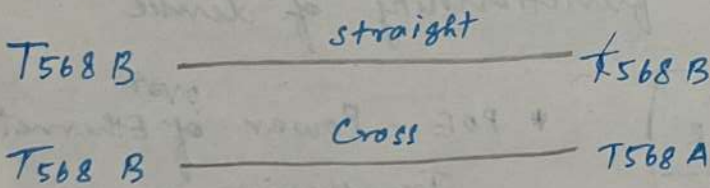
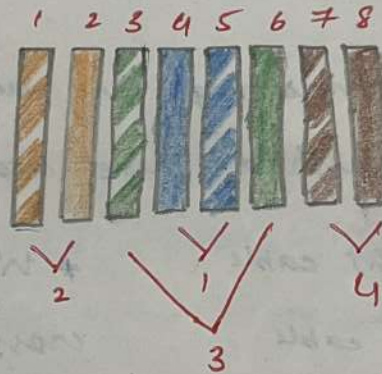
Different function device
Straight cable



T-568 A



T-568 B



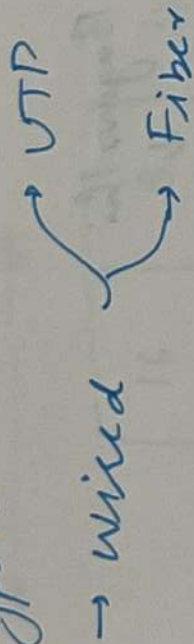
NIC:

- Network interface card
- It contains mac + ip.

Functions:

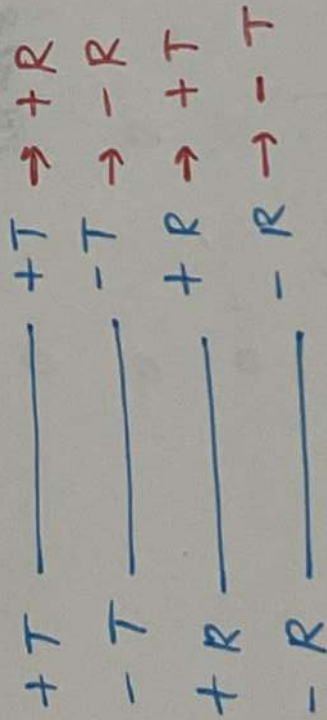
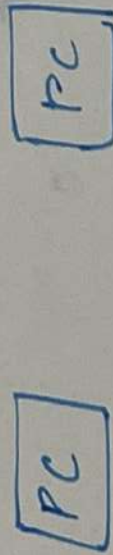
- Used to convert binary signals to electrical & electrical signals to binary

Types:



AUTO MDIX:

→ It can convert any cables into straight, even
or cross
it is a same functional device; Eg: PC & Router
or different functional device.



} → This is done by
AUTO MDIX.

Day-3 IP & MAC